Literature Review

Deep Learning for Mean-Field BSDEs with Jumps in Multi-Agent Dealer Markets

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April 2026

Abstract

This review establishes the probabilistic and computational foundations required to solve for Nash equilibria, and to detect tacit collusion, in the Cont–Xiong (2024) dealer market-making game via deep BSDE methods for jump-diffusion processes. Forty-two papers are organised across five sections — BSDE and mean-field game theory, the McKean–Vlasov FBSDE framework, deep-learning numerics, empirical dealer-market applications, and research frontier — and synthesised under a single arc: established knowledge, outstanding gaps, recent advances, and the motivation for the contributions of this thesis.

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1 Introduction

Multi-agent dealer market-making sits at an uncomfortable intersection in the mathematical literature. The N-player stochastic-differential-game formulation of Cont and Xiong [2024] admits, in principle, a probabilistic treatment via backward stochastic differential equations (BSDEs) and mean-field games (MFGs), and a numerical treatment via deep-learning methods. In practice, no existing framework simultaneously handles all three ingredients: Han, Hu, and Long [2022] give a curse-of-dimensionality-free deep-BSDE method for McKean–Vlasov FBSDEs but without jumps; Wang, Wang, Li, Wang, and Li [2023] extend deep BSDE to jump-diffusion processes but without mean-field coupling; Cont and Xiong [2024] solve their own game empirically via multi-agent reinforcement learning but without the equilibrium grounding or interpretable gradients that a principled method would supply. Combining all three — jump-augmented BSDEs, mean-field coupling, and deep-learning numerics — into a single computational method for a realistic market-microstructure model remains open, and the failure modes that emerge when one tries are not systematically documented in the literature.

This review is framed around a single target problem: the multi-agent dealer market-making game of Cont and Xiong [2024]. Every piece of theory surveyed in §2, every numerical tool catalogued in §3, and every empirical finding discussed in §4 is selected because it is needed to

solve that game. §2 assembles the probabilistic scaffolding — BSDEs, jump processes, mean-field limits, master equations — that lets us write down the Cont–Xiong value function. §3 covers the deep-learning numerical machinery that makes the resulting high-dimensional mean-field BSDE with jumps computable. §4 positions the thesis against the multi-agent reinforcement-learning literature in which Cont–Xiong was first solved empirically, and isolates what a principled BSDE treatment adds. §5 identifies what no existing work yet provides.

The starting point is Pardoux and Peng [1990], who established the existence and uniqueness of adapted solutions to BSDEs under Lipschitz conditions by a probabilistic fixed-point argument. Their result provides the theoretical bedrock on which every subsequent development rests: it guarantees that a BSDE, despite being specified "backwards" from a terminal condition, has a well-defined solution adapted to the forward filtration of a Brownian motion. The translation of this technology into quantitative finance was effected by Karoui, Peng, and Quenez [1997], who showed that pricing, hedging, and risk management under constraints admit a natural BSDE formulation. For a European contingent claim $\xi = g(S_T)$ in a Black–Scholes market with risk-free rate r, drift μ and volatility σ, the no-arbitrage price and replicating strategy (Y_t, Z_t) satisfy the linear BSDE

$$-dY_t \;=\; \big[rY_t + Z_t\theta\big]dt - Z_t\,dW_t, \qquad Y_T \;=\; g(S_T),$$

where $\theta = (\mu - r)/\sigma$ is the market price of risk and Z_t is the hedging portfolio in the risky asset. Their framework unified arbitrage-free pricing, superhedging via *reflected* BSDEs (for American options), and Epstein–Zin-type recursive utilities into a single probabilistic language, turning BSDEs from a purely theoretical object into a practical tool for derivative pricing and for the study of incomplete markets.

In parallel with the BSDE literature, Lasry and Lions — and independently Huang, Caines, and Malhamé — introduced mean-field games as a tractable approximation of N-player stochastic differential games in the regime $N \to \infty$. Carmona and Delarue [2013] recast this analytic theory in purely probabilistic terms, showing that an MFG equilibrium is characterised by a McKean–Vlasov forward-backward SDE system in which the distribution μ_t of a representative agent's state enters both the drift and the terminal condition; the stochastic maximum principle then supplies the natural optimality condition, avoiding the infinite-dimensional PDEs of the analytic formulation. Extensions have since covered games of optimal timing [Carmona, Delarue, and Lacker, 2017] and the long-time ergodic behaviour of MFG systems [Cardaliaguet, Lasry, Lions, and Porretta, 2012]; for a comprehensive survey of analytical and numerical methods, see Achdou, Cardaliaguet, Delarue, Porretta, and Santambrogio [2020].

2 Mathematical Framework

Any probabilistic treatment of BSDEs rests on Brownian motion, the Itô integral, and the classical theory of stochastic differential equations, standard references for which are Øksendal [1998]. Within this setting, a BSDE is a pair of adapted processes (Y_t, Z_t) satisfying

$$Y_t \;=\; \xi + \int_t^T f(s, Y_s, Z_s)\,ds - \int_t^T Z_s\,dW_s, \qquad t \in [0, T],$$

where ξ is an $\mathcal{F}_T$-measurable terminal condition and f is the driver. Pardoux and Peng [1990] proved that whenever f is uniformly Lipschitz in (y, z) with ξ and $f(\cdot, 0, 0)$ in $\mathbb{H}^2$ (the Hilbert space of square-integrable progressively measurable processes), there exists a unique adapted pair $(Y, Z) \in \mathbb{H}^2 \times \mathbb{H}^2$ solving the BSDE, obtained as the fixed point of the contraction

$$\Phi(Y, Z)_t \;:=\; \mathbb{E}\Big[\xi + \int_t^T f(s, Y_s, Z_s)\,ds \,\Big|\, \mathcal{F}_t\Big]$$

on the Banach space $\mathbb{H}^2 \times \mathbb{H}^2$. A large body of later work has explored the structural consequences of this theorem. The Pardoux–Peng existence theory has since been extended in several

directions relevant to this project. Feng, Wang, and Zhao [2018] established the weak-solution correspondence between quasi-linear PDEs and FBSDEs, which applies to the Cont–Xiong HJB system whose coefficients are quasi-linear rather than classical. Peng [2004] showed that BSDEs naturally generate *nonlinear expectations.* Given a driver $g(t, z)$ independent of y, the g-expectation of a terminal value $\xi \in L^2(\mathcal{F}_T)$ is defined as

$$\mathcal{E}^g[\xi] \;:=\; Y_0, \qquad \text{where } (Y, Z) \text{ solves the BSDE with driver } g \text{ and terminal } \xi.$$

When g is positively homogeneous and subadditive in z, the functional $\xi \mapsto \mathcal{E}^g[\xi]$ satisfies the Artzner–Delbaen–Eber–Heath axioms (monotonicity, translation invariance, positive homogeneity, subadditivity) of a coherent risk measure; convex g yields a convex risk measure in the sense of Föllmer–Schied. This elevates BSDE theory to an axiomatic foundation for risk measurement, placing the Z-process learned by later deep-BSDE algorithms in the role of a risk-sensitivity functional. The crucial martingale representation result underlying existence — that any square-integrable martingale adapted to a Brownian filtration admits a representation as a stochastic integral — is what makes the Z-process well-defined and is also what Han, E, and Jentzen will later exploit to parameterise Z by a neural network.

Mean-field BSDEs (MF-BSDEs) extend this picture by allowing the driver or the terminal condition to depend on the law of the solution itself. Buckdahn, Li, and Peng [2009b] established the connection between MF-BSDEs and non-local partial differential equations via viscosity solution theory: the solution of an MF-BSDE provides the unique viscosity solution of an associated non-local PDE on the state space. The non-locality reflects the fact that the distribution μ_t of the forward process enters the PDE, which immediately precludes any classical finite-difference numerical scheme above a few dimensions. A complementary paper [Buckdahn, Djehiche, Li, and Peng, 2009a] establishes the mean-field limit itself: starting from an N-player FBSDE system with empirical measure $\mu_t^N = N^{-1}\sum_{i=1}^N \delta_{X_t^{i,N}}$, the authors prove that each i-th particle converges in L^2 to a McKean–Vlasov BSDE with law μ_t at a rate

$$\sqrt{\mathbb{E}\sup_{t\le T}\left|X_t^{i,N} - X_t^i\right|^2 + \mathbb{E}\sup_{t\le T}\left|Y_t^{i,N} - Y_t^i\right|^2} \;\le\; \frac{C}{\sqrt{N}},$$

for a constant C depending on the Lipschitz constants of the drift, diffusion, and driver. This rate is of direct relevance to every empirical study of mean-field approximation at finite N: it gives a precise, non-asymptotic benchmark against which finite-N simulation error can be compared.

The jump-augmented case has a parallel classical theory that predates the deep-learning numerics discussed in §3. Tang and Li [1994] established necessary conditions for optimal control of stochastic systems with random jumps, extending Pontryagin's maximum principle to a setting in which execution events are discrete. Barles, Buckdahn, and Pardoux [1997] gave the BSDE/integro-partial differential equation correspondence that is the direct analogue of the Pardoux–Peng–Feynman–Kac bridge for Brownian BSDEs: the solution of a BSDE with a compensated Poisson jump term is the unique viscosity solution of an associated IPDE. Together these papers provide the probabilistic foundation on which any numerical jump-BSDE scheme — including the deep BSDEJ method of §3, and any MV-fictitious-play BSDEJ solver built on top of it — must ultimately rest.

The definitive probabilistic treatment of MFGs is the two-volume monograph of Carmona and Delarue [2018]. Volume I develops the mean-field FBSDE framework, the stochastic maximum principle, and well-posedness under structural conditions. Volume II extends the theory to mean-field games with common noise — where all agents are exposed to a shared stochastic shock, inducing conditional distributions $\mu_t(\omega^0)$ rather than deterministic measure flows — and to the master equation on the Wasserstein space. The master equation is a PDE in (t, x, μ) whose solution encodes the entire MFG value function, and its probabilistic study combines the Lions derivative with the FBSDE-flow machinery. Existence in the common-noise case was established independently by Carmona, Delarue, and Lacker [2016] using strong and weak formulations and

a Schauder fixed-point argument. Classical solutions of the master equation, and hence the differentiability of the MV-FBSDE flow, were obtained by Chassagneux, Crisan, and Delarue [2014] under strong regularity assumptions on the coefficients.

On the control-theoretic side, Bayraktar, Cosso, and Pham [2018] obtained a randomised dynamic programming principle and a nonlinear Feynman–Kac representation for McKean–Vlasov value functions, realising them as the Y-component of FBSDEs with autonomous forward processes. This result is the direct analogue, in the mean-field setting, of the classical Feynman–Kac bridge that motivates the deep BSDE method. Wu and Zhang [2020] sharpened the master-equation theory by introducing a genuine viscosity-solution notion for parabolic master equations on the Wasserstein space, with closed-loop controls, thereby removing some of the regularity restrictions that Chassagneux et al. needed.

Finer properties of the mean-field limit were studied by Delarue, Lacker, and Ramanan [2019], who established a central limit theorem for the N-player game fluctuations around their mean-field limit and characterised the Gaussian correction in terms of a linearised master equation. Their CLT result is particularly relevant for the present thesis because the Cont–Xiong market-making game is naturally formulated in a finite-N regime, and the $O(1/\sqrt{N})$ propagation of chaos of Buckdahn et al. [2009a] combined with the CLT corrections of Delarue et al. gives a quantitative picture of how finite-N Nash equilibria deviate from mean-field equilibria. Finally, Zhang [2023] studied stochastic recursive control problems of McKean–Vlasov type via the dynamic programming principle on an infinite-dimensional HJB equation, providing a further link between MV-FBSDE theory and the recursive-utility literature. Together, these results yield a mathematical framework that is well-posed, quantitatively understood at finite N, and equipped with a viscosity and a probabilistic representation — but still without a computable solution in any high-dimensional setting.

3 Numerical Methodology

The numerical analysis of BSDEs follows two historical phases. The first, beginning with Kloeden and Platen [1992], Kloeden, Platen, and Schurz [1994] for forward SDEs, establishes the discretisation benchmarks: for a step size Δt the Euler–Maruyama scheme has strong convergence order 1/2 (i.e. $\mathbb{E}|X_T - X_T^{\Delta t}| = O(\sqrt{\Delta t})$) and weak order 1, while the Milstein correction attains strong order 1 at the cost of requiring derivatives of the diffusion coefficient. Peng and Xu [2011] extended these ideas to BSDEs, developing backward time-stepping schemes — implicit Euler with rate $|Y_0^{\Delta t} - Y_0| = O(\sqrt{\Delta t})$ under standard smoothness and Crank–Nicolson with rate $O(\Delta t)$ — coupled with conditional-expectation estimators (quantisation, Malliavin weights, least-squares regression). These methods are rigorous and enjoy classical convergence rates, but they all share a fatal limitation: computing the conditional expectations $\mathbb{E}[Y_{t_{n+1}}|\mathcal{F}_{t_n}]$ by grid-based or regression methods scales as $O(M^d)$ in the state dimension d, so that $d \gtrsim 4{-}5$ is effectively out of reach. For the dealer-market problem of Cont and Xiong [2024] this renders the direct approach infeasible already at moderate numbers of competitors.

The second phase is inaugurated by E, Han, and Jentzen [2017] and its journal version Han, Jentzen, and E [2018], who reformulate a parabolic PDE as a BSDE and parameterise the gradient process Z by a deep neural network, with Y_0 being a single learnable scalar. The training objective is the mean-square deviation of the simulated terminal value Y_T^π from the prescribed $g(X_T)$. Once trained, the network encodes both the value function $u(0,x) = Y_0$ and its gradient $\nabla u(t,x) = Z_t/\sigma$ along all simulated paths. Empirically, this approach handles up to 100 dimensions on a laptop; the authors reported successful experiments on the Black–Scholes PDE with correlated assets, the Hamilton–Jacobi–Bellman equation for a linear-quadratic controller, and the Allen–Cahn PDE. The extension to fully nonlinear second-order BSDEs was carried out by Beck, E, and Jentzen [2019], who treated 100-dimensional Black–Scholes–Barenblatt, HJB, and G-Brownian-motion problems under the 2BSDE framework, with neural networks

approximating both the gradient and the Hessian. Both of these papers are empirical: rigorous convergence proofs for the original deep BSDE method are still absent, a gap that Carmona and Laurière [2019] partially fill by giving two algorithms for ergodic McKean–Vlasov control and proving convergence under network-architecture-dependent rates.

The step from BSDEs to McKean–Vlasov FBSDEs introduces a second source of difficulty: the coefficients now depend on the unknown law μ_t of the forward process, so the training loop must couple neural-network learning with a distribution-matching fixed-point iteration. Guo, Hu, Xu, and Zhang [2023] develop a general mean-field game (GMFG) framework in which the coupling passes through the joint state–action distribution (μ_t, ν_t) rather than the state distribution alone: rewards $r(x, a, \mu, \nu)$ and transition kernels $P(\cdot|x, a, \mu, \nu)$ depend on the empirical action law as well as the empirical state law, generalising classical MFGs to McKean–Vlasov type. They prove convergence of fictitious play to the GMFG Nash equilibrium under Lipschitz and contraction conditions on the associated Bellman operator, handling μ_t computationally through empirical particle representations. The decisive theoretical advance for the present thesis is Han et al. [2022], who propose a novel deep-learning algorithm for MV-FBSDEs with general mean-field interactions and prove that its convergence is *free of the curse of dimensionality* under monotonicity conditions. Their proof uses a class of integral probability metrics and a fictitious-play decomposition: the outer loop updates the mean-field measure using a population-averaged surrogate, and the inner loop solves a standard FBSDE by deep BSDE techniques. The complexity is shown to grow only polynomially with dimension, so long as the Lasry–Lions monotonicity condition holds on the coupling term. This monotonicity requirement is a real constraint in applications: for the Cont–Xiong coupling, global Lasry–Lions monotonicity on the Wasserstein space has not been established, and empirically monotonicity can be verified only along a one-dimensional slice parameterised by the mean competitor quote. The Han–Hu–Long convergence guarantee therefore applies only on the restricted slice of measure space relevant to a coupling like Cont–Xiong's, a subtlety that the original paper does not address. Modulo that caveat, the Han–Hu–Long algorithm is the natural template on which any MV-fictitious-play BSDEJ solver for the Cont–Xiong game would be built.

The extension of deep BSDE to jump-diffusion processes and forward BSDEs with jumps (FBSDEJs) is due to Wang et al. [2023], whose work is the computational counterpart of the classical BSDE/IPDE equivalence of Barles et al. [1997]. Wang and collaborators parameterise the jump coefficient U_t^a, U_t^b alongside Z_t by separate neural networks and report convergence in up to 100 dimensions. Two caveats warrant emphasis. First, as of writing, Wang et al. [2023] has accumulated only a handful of citations, making the deep BSDEJ numerical literature considerably less mature than the Brownian-only deep BSDE literature it extends; any downstream application must therefore do its own due diligence on convergence, on architectural ablations, and on the exact form of the compensated-martingale correction, which is subtle to implement correctly. Second, the Cont–Xiong model as currently formulated has discrete inventory $q_t \in \{-Q, \ldots, Q\}$, so the associated BSDE is pure-jump: the diffusion coefficient Z_t vanishes identically, and the network learns only (U_t^a, U_t^b). This is a degenerate case of the full deep BSDE method — the Z-gradient that Han et al. [2018] make the centrepiece of their method plays no role — and a continuous-inventory diffusion reformulation, in which q_t follows a stochastic-approximation SDE with $\sigma_q \neq 0$ and $Z_t = \sigma_q \partial_q V$ is learned genuinely, is a natural extension of the present work. Taken together, the deep BSDE literature up to 2023 provides every tool needed to solve the Cont–Xiong problem — Han–E–Jentzen for the Z-gradient, Han–Hu–Long for the MV distribution coupling, Wang et al. for the jumps, Barles–Buckdahn–Pardoux for the analytic foundation — but no existing study combines all three in the context of a realistic market microstructure model.

But "the tools exist" is not the same as "the method works out of the box". Practical deep BSDE solvers for mean-field games exhibit a cluster of failure modes that the optimistic original literature does not document. Three are particularly acute. *BatchNorm erasure*: when

population-level statistics (such as an aggregate competitor-quote summary m_t) are broadcast across a minibatch, batch normalisation collapses their variance to zero and the network learns a distribution-independent policy — the very feature that mean-field theory was supposed to capture is silently discarded. *Representation collapse under DeepSets*: the permutation-invariant pooling operators routinely used to process opponent states can, under symmetric initialisations, produce degenerate representations that ignore competitor-specific information. *Generator bypass*: when the network has enough capacity, it can satisfy the terminal-loss $|Y_T - g(q_T)|^2 \to 0$ with a U-profile that is internally inconsistent with the first-order-condition-derived optimal quotes, so that the solver converges to a self-consistent but non-Nash fixed point — a pathology that a naive MV-fictitious-play BSDEJ solver can exhibit. Germain, Pham, and Warin [2022] already catalogue vanishing gradients, terminal-scale sensitivity, and Y_0 basin-locking; the three failure modes listed above are specific to the *mean-field* extension and to the *jump-augmented* solver, and are not systematically addressed in the existing literature. These failures share a common structure: each of them eliminates or corrupts the policy's dependence on the population distribution — precisely the feature that distinguishes a mean-field problem from a single-agent one — so that the network satisfies the training loss without actually representing the mean-field interaction. Documenting and repairing them is an open problem that this thesis takes up.

4 Empirical Results and Financial Applications

The translation of deep BSDE and MV-FBSDE methods into finance has proceeded along several parallel streams, each reporting its own difficulties and partial successes. Germain et al. [2022] provides a critical survey of NN-based algorithms for stochastic control and PDEs in finance, documenting systematic training instabilities: vanishing gradients deep in the time discretisation, sensitivity to the terminal-loss scale, and a tendency for the Y_0 estimator to lock onto incorrect basins of attraction when the terminal condition is not smooth. Their paper is a useful antidote to the optimism of the original deep BSDE literature and motivates the careful regularisation and warm-start procedures that any practical deep-BSDE solver for a financial MFG must employ. Rigorous convergence rates for deep-NN stochastic control on finite horizons were obtained by Huré, Pham, Bachouch, and Langrené [2020], who combined deep RL with dynamic programming and proved error bounds depending on the regularity of the value function.

The pivot point of the empirical literature, for the purposes of the present thesis, is Han et al. [2022]. By giving a deep-learning algorithm for McKean–Vlasov FBSDEs whose convergence is free of the curse of dimensionality under monotonicity, Han, Hu, and Long unlocked the mean-field direction for deep BSDE. Their algorithm alternates between a fictitious-play outer loop, which updates the empirical measure of the population, and an inner deep-BSDE solve, which treats the measure as exogenous. Convergence is established in terms of the number of fictitious-play iterations and the supervised-learning error in the inner BSDE. The numerical experiments focus on an ergodic linear-quadratic MFG and a systemic risk model; the authors do not treat jump-driven games or discrete execution, leaving a clear gap that the Cont–Xiong application fills.

The Cont–Xiong model itself [Cont and Xiong, 2024] is the thesis's financial application blueprint. Cont and Xiong model N competing market makers in a continuous-time dealer market as a stochastic differential game of intensity control. Each market maker posts bid and ask quotes, and the execution probability in response to a random order flow depends on the maker's own quote relative to a softmax aggregate of competitors' quotes. Nash equilibrium is characterised by a coupled system of HJB equations (their equation 28), whose existence is established by Brouwer's fixed-point theorem, and whose computation Cont and Xiong reduce to a fictitious-play iteration that alternates policy evaluation (a linear Bellman solve) with a best-response step. The Pareto-optimum — which represents joint-objective collusion — is computed by a parallel iteration on the joint objective. Critically, Cont and Xiong then show

via decentralised multi-agent deep deterministic policy gradient (MADDPG) that competing market-making algorithms, learning independently from their own trade outcomes without any shared information, converge to spreads strictly above the Nash equilibrium, a phenomenon they term *tacit collusion*. This empirical finding has direct regulatory implications and reframes the MADDPG experiment as a detection mechanism: the gap between learned and Nash spreads is a measurable signature of algorithmic collusion even in the absence of explicit coordination.

A natural question follows. If Cont and Xiong have already solved their own model empirically via MADDPG, what does a BSDE-based treatment add? Three things. *Equilibrium grounding*: the BSDEJ solution directly targets the Nash fixed point of the coupled HJB system, not a behavioural attractor of a learning dynamic; the recovered object is an equilibrium by construction. *Interpretability*: the learned Z and U processes have direct economic meaning — the sensitivity of the value function to inventory and the jump in value at each execution event, respectively — whereas MADDPG's actor and critic networks are opaque. *Convergence structure*: Han–Hu–Long-style fictitious-play convergence, subject to the monotonicity caveat noted in §3, comes with quantitative error bounds; MADDPG offers none. The two methods are therefore complementary: MADDPG reveals what algorithms will actually do in practice, while BSDEJ establishes the benchmark from which they deviate. The *gap* between the two — quantified as $|S_{\text{MADDPG}} - S_{\text{Nash}}|$ — is the natural empirical object of interest for any BSDE-based treatment of the game.

Related MARL work on realistic LOBs has produced complementary results. Karpe, Fang, Ma, and Wang [2020] train multi-agent RL agents for optimal execution in the ABIDES simulator, showing that model-free RL can recover quasi-Almgren–Chriss policies when microstructure signals are available; their setup is close to the Cont–Xiong environment but lacks the stochastic-game structure. Shi and Cartlidge [2024] take a different angle, proposing a hybrid neural stochastic agent-based model in which a pre-trained neural point-process captures the background order-flow behaviour of uncontrolled traders, and an agent-based layer on top introduces strategic decisions. Both of these papers highlight a common empirical issue — generalisation across market regimes is fragile — that motivates the use of principled BSDE-based solvers, whose training only requires knowledge of the model coefficients rather than historical data.

What the empirical literature lacks, as of late 2025, is a deep BSDE-with-jumps solution of the Cont–Xiong MFG and a systematic finite-N to mean-field comparison. The closest existing attempt is the Han–Hu–Long algorithm applied to non-jump MFGs, combined with the Cont–Xiong Bellman-plus-MADDPG study on the finance side. Bridging these two is the technical aim of this thesis.

5 Conclusion and Open Problems

Three very recent papers define the research frontier and help to scope the contribution of the present thesis. Hu, Jin, and Laurière [2025] extend the MV-FBSDE machinery to environments with common noise by parameterising the measure flow through path signatures rather than empirical measures. Their deep signature approach combines signature features — a tractable compression of the history of a stochastic process — with neural networks that operate on the signature rather than on the full path, and establishes convergence in terms of fictitious-play iterations under supervised-learning error control. This is the natural mean-field generalisation of common-noise MFGs [Carmona et al., 2016, Carmona and Delarue, 2018] to the numerical regime and offers a promising route for future extensions of the present thesis to market making under systemic shocks.

Navarro-Rubio and Lage-Castellanos [2025] propose a unified mean-field framework that bridges informal bargaining markets and formal LOBs by modelling both as interacting-particle systems on a one-dimensional price lattice. Their paper supports the central narrative of this thesis — that dealer markets are natural mean-field environments — and suggests directions for

mapping the Cont–Xiong solver onto fully general LOB topologies. Finally, Cheridito, Dupret, and Wu [2025] present ABIDES-MARL, a GPU-accelerated multi-agent reinforcement learning environment built on top of ABIDES for studying endogenous price formation in LOBs. ABIDES-MARL addresses the scalability limitations of earlier LOB simulators by exploiting parallelism for both agent policy updates and environment simulation, providing a platform on which the tacit-collusion experiments of Cont and Xiong [2024] can in principle be extended to hundreds of agents.

Taken together, the literature gives a consistent diagnosis and a narrow research gap. BSDE theory and MFG theory are both mature, and their probabilistic synthesis via MV-FBSDEs is well understood [Pardoux and Peng, 1990, Carmona and Delarue, 2018, Buckdahn et al., 2009b, Chassagneux et al., 2014]. Deep-learning numerics for the Brownian-only case are well-studied, with both empirical validation [E et al., 2017, Han et al., 2018, Beck et al., 2019] and partial convergence theory [Carmona and Laurière, 2019, Huré et al., 2020], and have recently been extended to the curse-of-dimensionality-free MV setting [Han et al., 2022]. Financial applications to multi-agent market making are only beginning: Cont and Xiong [2024] provides the stochastic-game model and a decisive empirical finding on tacit collusion, and emerging frontiers include common-noise signatures [Hu et al., 2025], mean-field LOB embeddings [Navarro-Rubio and Lage-Castellanos, 2025], and GPU-scale MARL environments [Cheridito et al., 2025]. What no existing paper provides is, to state the central open question of this review in a single sentence: a validated BSDE-with-jumps solution of the Cont–Xiong game, an explanation of why naive deep BSDE methods fail in that setting, or a quantitative operationalisation of the MADDPG–Nash spread gap as a detector statistic for algorithmic collusion. These three absences motivate a single coherent research arc — *making the Cont–Xiong equilibrium computable, ensuring that the computation is correct, and extracting economically meaningful structure from the resulting solution* — which this thesis takes up through the following questions. (i) Can a deep BSDEJ solver recover the Cont–Xiong Nash equilibrium, and can the $O(1/\sqrt{N})$ propagation-of-chaos rate [Buckdahn et al., 2009a] be verified empirically on this specific model? (ii) Which failure modes (BatchNorm erasure, representation collapse, generator bypass) derail naive applications of deep BSDE to mean-field games, and how can they be systematically diagnosed and repaired? (iii) Can the spread gap $|S_{\mathrm{MADDPG}} - S_{\mathrm{Nash}}|$ be elevated from an empirical observation in Cont and Xiong [2024] to a measurement tool for regulators of algorithmic markets? Taken together, these three questions define the scope of the thesis that this review sets up.

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